Source-energy selection and perihelion

From SET capacity to the spherical field and Mercury's orbital advance

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The question is what makes SET choose the gravitational response before we calculate an orbit. The principle used here is that the energy establishing a source's gravitational charge must agree with the complete energy of that same source: its matter, field and any stresses supporting it.

I leave both the clock response and the radial response unsolved. Their energy difference can then be written as an integral of nonnegative terms. Exact agreement with the source energy forces those terms to vanish. The selected field gives the perihelion equation, with the source law still visible in the result.

This derivation uses the capacity relation, the source law, a common canonical energy for motion and clocks, and the static field-energy functional displayed below. The field is isolated, stationary and spherical, with a positive clock rate outside the source. Each retained relation is stated before it is used.

1. Capacity: motion and clocks share the same limit

In SET, mass continuously commits field interaction capacity as gravitational burden. Translation and local physical evolution participate in the remaining capacity. The starting relation is

$$S^2 + V_{\mathrm{rel}}^2 + c^2\beta^2 = c^2. \tag{1}$$

Here S and V_{rel} have speed units, T is the reference time and $\beta = d\tau/dT$ is the moving clock's progression. Define the held-clock factor α and its square A:

$$A = \alpha^2 = 1 - \frac{S^2}{c^2}, \qquad \beta^2 = A - \frac{V_{\mathrm{rel}}^2}{c^2}. \tag{2}$$

Multiplying the invariant by $m/2$ exposes its exact quadratic balance:

$$\begin{gathered} \frac{mS^2}{2} + K_0 + \frac{mc^2\beta^2}{2} = \frac{mc^2}{2}, \\ K_0 = \frac{mV_{\mathrm{rel}}^2}{2}, \qquad dK_0 = -\frac{m}{2}\,dS^2 - \frac{mc^2}{2}\,d\beta^2. \end{gathered} \tag{3}$$

Burden, translation and clock progression cannot change independently. To determine an actual orbit, we now require the same energy to generate its motion and its internal clock.

2. The source law fixes the charge and source boundary

Let E_s be the complete source energy and $M = E_s/c^2$ its mass equivalent. For a specified source domain, SET's law is

$$Q_s^2 = 24\pi G \frac{E_s}{c^2} V_{\mathrm{ref}}, \qquad \mu_Q \equiv \frac{Q_s^2}{24\pi V_{\mathrm{ref}}} = \frac{GE_s}{c^2} = GM. \tag{4}$$

For a spherical source of fixed areal radius R_0, use $V_{\mathrm{ref}} = 4\pi R_0^3/3$. This is the specified source-law volume. The physical source surface has area $4\pi R_0^2$, so

$$Q_s = 4\pi R_0^2 S_0, \qquad Q_s^2 = 32\pi^2 \mu_Q R_0^3, \qquad S_0 = S(R_0). \tag{5}$$

Squaring the surface reading and comparing it with the source law gives

$$\begin{aligned} S_0^2 &= \frac{2\mu_Q}{R_0}, \qquad b \equiv \frac{\mu_Q}{c^2} = \frac{GE_s}{c^4}, \\ A_0 \equiv A(R_0) &= 1 - \frac{2b}{R_0}, \qquad \alpha_0 = \sqrt{A_0}\,. \end{aligned} \tag{6}$$

The source radius R_0 and exterior evaluation radius r have different jobs. Q_s belongs to the fixed source domain. At this point the law has fixed its charge and surface clock reading; the exterior functions remain to be selected.

3. One energy generates the probe's motion and clock

Let q_{ij} specify the physical spatial reading and $v^i = dx^i/dT$. In the stationary description used here there is no transport shift:

$$V_{\mathrm{rel}}^2 = q_{ij} v^i v^j, \qquad \beta^2 = A - \frac{q_{ij} v^i v^j}{c^2}. \tag{7}$$

Retain regular canonical evolution for a test body. Its reference energy is $E_p(x, p, E_0)$, where E_0 is its local rest energy. The same energy generates translation and internal evolution.

To make the clock condition explicit, let an internal phase θ have conjugate action I and internal energy $h(I)$, with $E_0 = E_{\mathrm{base}} + h(I)$. Its canonical and clock readings are

$$\frac{d\theta}{dT} = \frac{\partial E_p}{\partial E_0} h'(I), \qquad \frac{d\theta}{dT} = \beta h'(I). \tag{8}$$

For a functioning phase, $h'(I) \neq 0$. The two readings therefore require

$$\begin{aligned} v^i &= \frac{\partial E_p}{\partial p_i}, \qquad \frac{\partial E_p}{\partial E_0} = \beta, \\ q_{ij} \frac{\partial E_p}{\partial p_i} \frac{\partial E_p}{\partial p_j} &+ c^2 \left(\frac{\partial E_p}{\partial E_0} \right)^2 = c^2 A. \end{aligned} \tag{9}$$

The derivative in rest energy compares neighboring internal states at fixed momentum. The body can retain the same rest energy throughout its orbit.

At test-body order, retain additive scaling in a supplied field. Scaling the body's rest energy and momentum together scales its energy by the same amount. Euler's identity then gives

$$\begin{gathered} E_p(x, \lambda p, \lambda E_0) = \lambda E_p(x, p, E_0), \qquad \lambda > 0, \\ E_p = E_0\beta + p_i v^i. \end{gathered} \tag{10}$$

Differentiate the second line with respect to momentum. The direct velocity terms cancel:

$$0 = \left(p_i + E_0 \frac{\partial \beta}{\partial v^i}\right)\frac{\partial v^i}{\partial p_j}, \qquad \frac{\partial \beta}{\partial v^i} = -\frac{q_{ik}v^k}{c^2\beta}. \tag{11}$$

On the regular positive massive branch, the momentum Hessian is invertible. The bracket must vanish. Substitution back into the common energy gives

$$\begin{gathered} p_i = \frac{E_0 q_{ij} v^j}{c^2\beta}, \qquad E_p = E_0\left(\beta + \frac{q_{ij}v^i v^j}{c^2\beta}\right) = \frac{E_0 A}{\beta}, \\ E_p = \alpha\sqrt{E_0^2 + c^2 q^{ij} p_i p_j}\,. \end{gathered} \tag{12}$$

At fixed internal action and E_0, the Legendre transform now fixes the reduced probe Lagrangian:

$$L_p = p_i v^i - E_p = -E_0\beta, \qquad \mathcal{I}_p = -mc^2 \int \beta \, dT = -mc^2 \int d\tau. \tag{13}$$

Here $m = E_0/c^2$, as identified by the unloaded low-speed momentum. The clock term has become the probe action through the common-energy requirement. In the weak field the same energy gives

$$\begin{gathered} E_p - mc^2 = \frac{mV_{\mathrm{rel}}^2}{2} - \frac{mS^2}{2} + O\left(\frac{m(S^2 + V_{\mathrm{rel}}^2)^2}{c^2}\right), \\ \Phi = -\frac{S^2}{2}, \qquad \mathbf{a} = \frac{1}{2}\nabla S^2. \end{gathered} \tag{14}$$

The attractive potential is already visible. The full orbital correction also depends on the radial response, which is the next part of the selection.

4. The field principle and the energy being compared

Write the stationary spherical spatial reading as

$$d\ell^2 = B(r)\,dr^2 + r^2 d\Omega^2, \qquad d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2,$$
$$\alpha = \sqrt{A}, \qquad \kappa = \sqrt{AB} > 0. \tag{15}$$

Both A and B are unknown. We compare smooth, positive exterior profiles with finite combined field energy and $r \geq R_0 > 2b$. The reference clock is normalized by $\alpha(\infty) = 1$. The isolated spatial energy is normalized to the same source charge:

$$B(r) = 1 + \frac{2b}{r} + O(r^{-2}). \tag{16}$$

The static part of the retained common-clock field construction supplies the following complete energy. In isotropic spatial coordinates, write

$$q_{ij} = \psi^4 \delta_{ij}, \qquad dV = \psi^6 d^3x,$$
$$W_f = -\frac{c^4}{2\pi G}\int \nabla\psi \cdot \nabla(\alpha\psi)\, d^3x, \qquad W_m = \int \alpha\varepsilon\, dV. \tag{17}$$

Here ε is proper material and support energy density. The displayed gradients use the isotropic Euclidean coordinates. This is the field-energy functional used for every candidate, before a field solution is inserted. Together with the common canonical framework and source conditions, it specifies the class of this proof.

Stationary lapse and spatial variations of this functional give the pair developed in Appendix A:

$$\Delta\psi = -\frac{2\pi G}{c^4}\psi^5\varepsilon, \qquad \Delta(\alpha\psi) = \frac{2\pi G}{c^4}\alpha\psi^5(\varepsilon + 2\mathrm{tr}\sigma). \tag{18}$$

The stress σ includes the physical supports. In physical spatial lengths these equations give a work-current and its source:

$$\mathbf{J}_W = c^2 D\alpha = -\frac{D(S^2)}{2\alpha},$$
$$D\cdot\mathbf{J}_W = \frac{4\pi G}{c^2}\alpha(\varepsilon + \mathrm{tr}\sigma). \tag{19}$$

The complete stationary energy accounting identifies $E_s = \int\alpha(\varepsilon + \mathrm{tr}\sigma)dV$. Integrating the current over a complete vacuum sphere therefore reads the source charge:

$$\int_{\Sigma_r}\mathbf{J}_W \cdot d\mathbf{A} = \frac{4\pi G E_s}{c^2} = 4\pi\mu_Q = \frac{Q_s^2}{6V_{\mathrm{ref}}},$$
$$J_W = \frac{c^2\alpha'}{\sqrt{B}}, \qquad r^2 J_W = c^2 b. \tag{20}$$

Primes here mean radial derivatives. With $\kappa = \sqrt{AB}$ this becomes

$$A' = \frac{2b\kappa}{r^2}. \tag{21}$$

This makes A monotone, but leaves κ unsolved. The global comparison below retains the current, boundary data and energy functional while allowing different radial profiles. The local spatial equation is not imposed on those competing profiles.

5. Fix the complete source budget independently of B

Use a supported thin spherical shell of fixed physical radius R_0 to make the energy comparison explicit. Let E_m include its proper material and support energy. Its interior fields are constant, so its field-gradient energy is entirely exterior.

The same common-energy clock condition is retained when the complete supported source is loaded and its field readjusts. Combining it with the source-surface law gives

$$g \equiv \frac{G}{R_0 c^4}, \qquad \frac{\partial E_s}{\partial E_m} = \alpha_0 = \sqrt{1 - 2gE_s}, \qquad E_s(0) = 0. \tag{22}$$

Set $z = \sqrt{1 - 2gE_s}$. Differentiation removes the square root:

$$\begin{gathered} \frac{dz}{dE_m} = -\frac{g}{z}\frac{dE_s}{dE_m} = -g, \qquad z(0) = 1, \\ \alpha_0 = 1 - gE_m, \qquad E_s = E_m - \frac{gE_m^2}{2}, \qquad 0 < gE_m < 1. \end{gathered} \tag{23}$$

The material contribution in reference-energy units is $\alpha_0 E_m$. The field must supply the remainder:

$$\begin{gathered} W_{\text{req}} \equiv E_s - \alpha_0 E_m = \frac{gE_m^2}{2} = E_s \frac{1 - \alpha_0}{1 + \alpha_0}, \\ W_{\text{total}} = \alpha_0 E_m + W_{\text{ext}}. \end{gathered} \tag{24}$$

This budget contains no choice of B and no orbital observation. Each candidate must describe this same source and support state.

6. Evaluate the field energy before selecting its profile

Let $f = B^{-1/2}$ and use isotropic radius ρ. Matching the angular and radial spatial readings gives

$$r = \rho\psi^2, \qquad f = 1 + \frac{2\rho\psi_\rho}{\psi}, \qquad dr = \psi^2 f\, d\rho. \tag{25}$$

The angular integral of the field functional is

$$W_{\text{ext}} = -\frac{2c^4}{G}\int_{\rho_0}^{\rho_c} \rho^2 \psi_\rho (\alpha\psi)_\rho \, d\rho. \tag{26}$$

Use the two derivatives below and change the integration variable to the physical areal radius:

$$\psi_\rho = \frac{\psi(f-1)}{2\rho}, \qquad \alpha_\rho = \alpha' \psi^2 f. \tag{27}$$

$$W_{\text{ext}} = -\frac{c^4}{2G}\int_{R_0}^{r_c}\left[\frac{\alpha(f-1)^2}{f} + 2r\alpha'(f-1)\right]dr. \tag{28}$$

The current supplies $\alpha' f = b/r^2$, while $f = \alpha/\kappa$. Expanding the integrand gives an identity valid before selection:

$$\frac{\alpha(f-1)^2}{f} + 2r\alpha'(f-1) = \frac{A}{\kappa} + \kappa + \frac{2b}{r} - 2(r\alpha)'. \tag{29}$$

We keep a finite outer cutoff because the combined energy is finite even when separately integrated pieces are not.

First test one comparison profile:

$$A_*(r) = 1 - \frac{2b}{r}, \qquad B_*(r) = A_*^{-1}. \tag{30}$$

For this profile $f = \alpha$ and $r = 2b/(1-\alpha^2)$. Direct substitution into the full radial energy gives

$$W_* = 2E_s\int_{\alpha_0}^{1}\frac{d\alpha}{(1+\alpha)^2} = E_s\frac{1-\alpha_0}{1+\alpha_0} = W_{\text{req}}. \tag{31}$$

One profile closes the budget. The next step determines whether another admissible profile can close it too.

7. Turn every deviation into an exact energy gap

Since A is monotone, use $a = A(r)$ as the integration variable. This a is only a capacity variable. Define

$$\begin{gathered} y(a) = \ln\frac{2b}{r(a)}, \qquad y_*(a) = \ln(1-a), \\ \eta(a) = y(a) - y_*(a) = \ln\frac{2b}{r(a)(1-a)}. \end{gathered} \tag{32}$$

The comparison profile has $\eta = 0$. A subscript a now denotes differentiation with respect to this new variable. The current and change of variable give

$$\frac{dr}{da} = -ry_a, \qquad \kappa = -\frac{e^{-y}}{y_a}. \tag{33}$$

The first two terms of the energy integrand transform as

$$F[y] \equiv \int\left(\frac{A}{\kappa} + \kappa\right)dr = 2b\int\left(ay_a^2 + e^{-2y}\right)da. \tag{34}$$

Compare the two profiles at a common upper capacity $a_c < 1$. Expanding $y = y_* + \eta$ gives

$$F[y] - F[y_*] = 2b\int_{A_0}^{a_c}\left[a\eta_a^2 + 2ay_{*a}\eta_a + e^{-2y_*}(e^{-2\eta} - 1)\right]da. \tag{35}$$

The comparison function has the exact derivative identity

$$\frac{d}{da}(ay_{*a}) = -\frac{1}{(1-a)^2} = -e^{-2y_*}. \tag{36}$$

Integrate the cross term once. Its interior part supplies the linear term needed to complete a nonnegative expression:

$$\begin{aligned} F[y] - F[y_*] &= 2bI_c[\eta] + 4b[ay_{*a}\eta]_{A_0}^{a_c}, \\ I_c[\eta] &= \int_{A_0}^{a_c}\left[a\eta_a^2 + \frac{e^{-2\eta} - 1 + 2\eta}{(1-a)^2}\right]da. \end{aligned} \tag{37}$$

The logarithmic and derivative terms in (29) contribute endpoints as well. Keeping all of them together gives

$$\begin{aligned} W_{\mathrm{ext},\,c} - W_{*,\,c} &= -\frac{c^4}{2G}(2bI_c[\eta] + \mathcal{B}(a_c) - \mathcal{B}(A_0)), \\ \mathcal{B}(a) &= 4bay_{*a}\eta - 2b\eta - 2\sqrt{a}\,(r - r_*). \end{aligned} \tag{38}$$

The fixed physical source surface gives $\eta(A_0) = 0$. The isolated energy normalization and current give $\eta = O((1-a)^2)$ at infinity. The *combined* endpoint therefore vanishes at both ends. Appendix B shows this limit explicitly.

Use $W_* = E_s - \alpha_0 E_m$ and $c^4 b/G = E_s$. The complete energy comparison is

$$\frac{W_{\mathrm{total}} - E_s}{E_s} = -\int_{A_0}^{1}\left[a\left(\frac{d\eta}{da}\right)^2 + \frac{e^{-2\eta} - 1 + 2\eta}{(1-a)^2}\right]da. \tag{39}$$

8. Energy agreement selects both A and B

Every term inside the integral is nonnegative:

$$a > 0, \qquad a\eta_a^2 \geq 0, \qquad e^{-2\eta} - 1 + 2\eta \geq 0. \tag{40}$$

The last inequality is the convexity of the exponential. Equality holds only at $\eta = 0$. Therefore the required complete-source agreement has a definite consequence:

$$W_{\text{total}} = E_s \quad \Longrightarrow \quad \eta(a) = 0 \quad \Longrightarrow \quad r(1-A) = 2b. \tag{41}$$

Differentiate the last expression and compare it with the current (21). This fixes κ as well:

$$\begin{aligned} A &= 1 - \tfrac{2b}{r}, \qquad A' = \tfrac{2b}{r^2} = \tfrac{2b\kappa}{r^2}, \\ \kappa &= 1, \qquad AB = 1, \qquad B = A^{-1}. \end{aligned} \tag{42}$$

This is the selection result. Both the burden profile and radial response follow from exact energy agreement within the stated static field class. The candidates were not assigned $AB = 1$, and no perihelion value was used.

The condition is energy consistency. The proof does not require an additional instruction to maximize the energy.

9. The same exterior applies outside a spherical star

The shell makes the global comparison transparent. For a star, use its actual interior and support energy and the local stationary equation of the same functional (18). The conformal spatial identity gives

$$\mathcal{R}[q] = -8\psi^{-5}\Delta\psi = \tfrac{16\pi G}{c^4}\varepsilon. \tag{43}$$

For $d\ell^2 = B\,dr^2 + r^2 d\Omega^2$, its spherical form is

$$\mathcal{R}[q] = \tfrac{2}{r^2}\tfrac{d}{dr}\left[r(1 - B^{-1})\right]. \tag{44}$$

In the exterior material and support vacuum, the bracket is constant. The same isolated source-energy normalization fixes that constant:

$$r(1 - B^{-1}) = 2b, \qquad B = \left(1 - \tfrac{2b}{r}\right)^{-1}. \tag{45}$$

The conserved work-current then determines the clock with its infinity normalization:

$$\begin{aligned} \alpha' &= \tfrac{b}{r^2\sqrt{1 - 2b/r}}, \qquad \alpha(\infty) = 1, \\ \alpha &= \sqrt{1 - \tfrac{2b}{r}}, \qquad A = 1 - \tfrac{2b}{r}. \end{aligned} \tag{46}$$

Thus the same exterior follows for an admissible spherical star. This is the step that lets the orbital calculation use the Sun's complete source energy.

10. Read the selected field in SET source quantities

The selected responses can now be written entirely in the fixed source-domain variables:

$$\mu_Q = \frac{Q_s^2}{32\pi^2 R_0^3}, \qquad S^2(r) = \frac{2\mu_Q}{r} = \frac{Q_s^2}{16\pi^2 R_0^3 r},$$
$$A(r) = 1 - \frac{Q_s^2}{16\pi^2 R_0^3 c^2 r}, \qquad B(r) = \frac{1}{A(r)}. \tag{47}$$

The inverse-area quantity in this exterior is the physical work-current:

$$J_W = \frac{\mu_Q}{r^2} = -\frac{1}{2}\frac{dS^2}{dr}, \qquad r^2 J_W = \mu_Q. \tag{48}$$

The exterior screen readout is a different quantity from the fixed source-domain Q_s:

$$Q_{\mathrm{scr}}(r) = 4\pi r^2 S(r) = Q_s \left(\frac{r}{R_0}\right)^{3/2},$$
$$\frac{Q_{\mathrm{scr}}^2(r)}{24\pi(4\pi r^3/3)} = \mu_Q. \tag{49}$$

The same source charge is read at every exterior screen. This distinction preserves the source radius and measurement radius without replacing the selected burden by an inverse-area speed.

Multiplying the capacity relation by the squared reference-time increment gives the operational interval:

$$c^2 d\tau^2 = A c^2 dT^2 - A^{-1} dr^2 - r^2 d\Omega^2. \tag{50}$$

This is the Schwarzschild interval. In this derivation its clock and radial coefficients have been selected through the displayed source-energy construction.

11. Conserved energy gives the exact orbital equation

Use the equatorial plane and write dots for d/dT. The selected probe Lagrangian is

$$L_p = -mc^2\beta, \qquad \beta^2 = A - \frac{B\dot{r}^2 + r^2\dot{\varphi}^2}{c^2}. \tag{51}$$

Stationarity and rotational symmetry give the two orbital constants:

$$\varepsilon \equiv \frac{E_p}{mc^2} = \frac{A}{\beta}, \qquad j \equiv \frac{p_\varphi}{m} = r^2\frac{d\varphi}{d\tau}. \tag{52}$$

Here j is angular momentum per unit probe mass. Since $dr/dT = \beta\, dr/d\tau$, division of capacity by β^2 gives

$$c^2 = \frac{Ac^2}{\beta^2} - B\left(\frac{dr}{d\tau}\right)^2 - \frac{j^2}{r^2}. \tag{53}$$

Multiply by A and use $A/\beta = \varepsilon$:

$$AB\left(\frac{dr}{d\tau}\right)^2 + A\left(c^2 + \frac{j^2}{r^2}\right) = c^2\varepsilon^2. \tag{54}$$

Insert the selected $AB = 1$. For $w = 1/r$ and primes now meaning $d/d\varphi$,

$$\begin{aligned} \frac{dr}{d\tau} &= -jw', \\ j^2(w')^2 &= c^2\varepsilon^2 - \left(1 - \frac{2\mu_Q w}{c^2}\right)(c^2 + j^2w^2). \end{aligned} \tag{55}$$

Expand the product before differentiating:

$$j^2(w')^2 = c^2(\varepsilon^2 - 1) + 2\mu_Q w - j^2w^2 + \frac{2\mu_Q j^2 w^3}{c^2}. \tag{56}$$

Differentiation with respect to φ gives

$$2j^2 w' w'' = 2\mu_Q w' - 2j^2 w w' + \frac{6\mu_Q j^2 w^2 w'}{c^2}. \tag{57}$$

Divide by $2j^2w'$ where it is nonzero. The smooth equation extends to an apsis; Appendix C verifies it there directly from the radial motion equation.

$$\begin{aligned} \frac{d^2w}{d\varphi^2} + w &= \frac{\mu_Q}{j^2} + \frac{3\mu_Q w^2}{c^2} \\ &= \frac{Q_s^2}{32\pi^2 R_0^3 j^2} + \frac{3Q_s^2 w^2}{32\pi^2 R_0^3 c^2}. \end{aligned} \tag{58}$$

The coefficient three comes from the cubic term. That term was generated by the selected burden response multiplying the angular-motion contribution. It was not adjusted to an orbit.

12. Extract the perihelion advance

Define the gravitational length and leading orbital scale:

$$\ell_g = \frac{\mu_Q}{c^2}, \qquad p_0 = \frac{j^2}{\mu_Q}, \qquad w_0 = \frac{1 + e\cos\varphi}{p_0}. \tag{59}$$

The Newtonian part satisfies $w_0'' + w_0 = 1/p_0$. Write $w = w_0 + \delta w$ and keep first order in ℓ_g/p_0:

$$\begin{aligned} \delta w'' + \delta w &= 3\ell_g w_0^2 \\ &= \frac{3\ell_g}{p_0^2}\left(1 + \frac{e^2}{2} + 2e\cos\varphi + \frac{e^2}{2}\cos 2\varphi\right). \end{aligned} \tag{60}$$

The constant and double-frequency terms produce bounded changes. The cosine at the original orbital frequency gives the accumulating phase shift. Since

$$\begin{aligned} \left(\frac{d^2}{d\varphi^2} + 1\right)(\varphi\sin\varphi) &= 2\cos\varphi, \\ \delta w_{\mathrm{res}} &= \frac{3\ell_g e}{p_0^2}\varphi\sin\varphi. \end{aligned} \tag{61}$$

Compare it with a slightly shifted radial oscillation:

$$\frac{e}{p_0}\cos[(1-\delta)\varphi] = \frac{e}{p_0}\cos\varphi + \frac{e\delta}{p_0}\varphi\sin\varphi + O(\delta^2). \tag{62}$$

The coefficients agree when

$$\delta = \frac{3\ell_g}{p_0}, \qquad \Delta\varpi = \frac{2\pi}{1-\delta} - 2\pi = \frac{6\pi\mu_Q}{p_0 c^2} + O\left(\frac{\mu_Q^2}{p_0^2 c^4}\right). \tag{63}$$

At this order $p_0 = a_{\mathrm{orb}}(1-e^2)$, with a_{orb} the semimajor axis. The advance per radial period is therefore

$$\begin{aligned} \Delta\varpi &\simeq \frac{6\pi GM}{a_{\mathrm{orb}}(1-e^2)c^2} \\ &= \frac{3Q_s^2}{16\pi R_0^3 a_{\mathrm{orb}}(1-e^2)c^2}. \end{aligned} \tag{64}$$

The source law, clock and radial selection, and probe dynamics have now reached the perihelion formula. The nonclosing radial cycle follows from their equations; it was not introduced as a geometrical correction to an ellipse.

13. Mercury: substitute the measured orbital inputs

Use the same benchmark inputs recorded in Integrated Rough Draft 6:

$$\begin{aligned} \mu_\odot &= 1.3271244004127942\times10^{20}\ \mathrm{m^3\,s^{-2}}, \qquad c = 299792458\ \mathrm{m\,s^{-1}}, \\ a_{\mathrm{orb}} &= 0.38709927\ \mathrm{au}, \qquad e = 0.20563593, \qquad 1\ \mathrm{au} = 149597870700\ \mathrm{m}. \end{aligned} \tag{65}$$

These give

$$\begin{aligned} p_0 \simeq a_{\mathrm{orb}}(1-e^2) &= 5.546046913\times10^{10}\ \mathrm{m}, \\ \tfrac{\mu_\odot}{p_0c^2} &= 2.662482056\times10^{-8}. \end{aligned} \tag{66}$$

Substitution into (64) gives the advance per orbit:

$$\Delta\varpi = 5.018660440\times10^{-7}\ \mathrm{rad} = 0.1035173023\ \mathrm{arcsec}. \tag{67}$$

With 36,525 days per Julian century and an orbital period of 87.969 days,

$$\Delta\varpi_{\mathrm{century}} = 0.1035173023\ \tfrac{36525}{87.969}\ \mathrm{arcsec} \simeq 42.98\ \mathrm{arcsec}. \tag{68}$$

This is the isolated spherical relativistic contribution. In source variables the same solar input is $Q_s^2/R_0^3 = 32\pi^2\mu_\odot$. Its normalization and the response coefficients were fixed before Mercury's orbital elements were substituted.

What the proof establishes: within the displayed stationary spherical construction, complete source-energy agreement removes the competing responses and selects the Schwarzschild clock and radial relationship. The same common energy fixes the probe action. Their combination yields the exact orbit equation and its leading perihelion advance.

Appendix A. Where the stationary current comes from

For completeness, vary the retained static energy at fixed canonical material and spatial data. A compact lapse variation gives

$$\delta_\alpha W_f = \tfrac{c^4}{2\pi G}\int \psi\Delta\psi\,\delta\alpha\,d^3x, \qquad \delta_\alpha W_m = \int \psi^6\varepsilon\,\delta\alpha\,d^3x. \tag{69}$$

Stationarity for arbitrary compact variations gives the first equation of (18). For a conformal spatial variation, the material stress response at fixed canonical material data is

$$\begin{aligned} \delta_\psi W_m &= -2\int \alpha\psi^5 \mathrm{tr}\sigma\,\delta\psi\,d^3x. \\ \delta_\psi W_f &= \tfrac{c^4}{2\pi G}\int [\Delta(\alpha\psi) + \alpha\Delta\psi]\,\delta\psi\,d^3x. \end{aligned} \tag{70}$$

Set the sum to zero and insert the lapse equation. This gives the second equation of (18). The physical spatial Laplacian is

$$\begin{aligned} D^2\alpha &= \psi^{-6}\nabla\cdot(\psi^2\nabla\alpha) \\ &= \psi^{-6}[\psi\Delta(\alpha\psi)-\alpha\psi\Delta\psi] = \tfrac{4\pi G}{c^4}\alpha(\varepsilon+\mathrm{tr}\sigma). \end{aligned} \tag{71}$$

Multiplication by c^2 produces the work-current law (19). The current and complete source-energy normalization therefore belong to the same static construction.

In Integrated Rough Draft 6 this functional is the static part of the retained common-lapse, common-spatial-reading canonical field system. That field completion has the ADM gravitational form. The proof here uses its stated energy functional; the capacity interpretation and source law do not by themselves replace that field premise.

Appendix B. Keep the boundary terms until they cancel

At equal capacity a, the two coordinate radii are $r = 2be^{-\eta}/(1-a)$ and $r_* = 2b/(1-a)$. From (29), the logarithmic term contributes $-2b\eta$ and the final radial derivative contributes $-2\sqrt{a}\,(r-r_*)$. Together with the cross term,

$$\begin{aligned} \mathcal{B}(a) &= 4bay_{*a}\eta - 2b\eta - 2\sqrt{a}\,(r-r_*) \\ &= -\tfrac{2b}{1-a}\left[(1-\sqrt{a})^2\eta + 2\sqrt{a}\,(e^{-\eta}-1+\eta)\right]. \end{aligned} \tag{72}$$

At the source, $A_0 = 1-2b/R_0$ implies $\eta(A_0)=0$. For the outer endpoint, combine the falloff (16) with the current:

$$\begin{aligned} \alpha' &= \tfrac{b\sqrt{B}}{r^2} = \tfrac{b}{r^2} + \tfrac{b^2}{r^3} + O(r^{-4}), \\ \alpha &= 1 - \tfrac{b}{r} - \tfrac{b^2}{2r^2} + O(r^{-3}), \qquad A = 1 - \tfrac{2b}{r} + O(r^{-3}). \end{aligned} \tag{73}$$

It follows that

$$\eta = \ln\tfrac{2b}{r(1-A)} = O(r^{-2}) = O((1-a)^2). \tag{74}$$

Since $(1-\sqrt{a})^2 = O((1-a)^2)$ and $e^{-\eta}-1+\eta = O(\eta^2)$, the combined endpoint tends to zero. This is the limit used to obtain (39).

Appendix C. Check the orbit at its turning points

The radial Euler-Lagrange equation obtained from (51), with primes here denoting d/dr, is

$$B\frac{d^2r}{d\tau^2} + \frac{B'}{2}\left(\frac{dr}{d\tau}\right)^2 - r\left(\frac{d\varphi}{d\tau}\right)^2 = -\frac{c^2\varepsilon^2A'}{2A^2}. \tag{75}$$

Use $B = 1/A$ and the radial energy equation (54). The acceleration becomes

$$\begin{aligned}\frac{d^2r}{d\tau^2} &= -\frac{A'}{2}\left(c^2 + \frac{j^2}{r^2}\right) + \frac{Aj^2}{r^3} \\ &= -\frac{\mu_Q}{r^2} + \frac{j^2}{r^3} - \frac{3\mu_Q j^2}{c^2r^4}.\end{aligned} \tag{76}$$

For $w = 1/r$, $d^2r/d\tau^2 = -j^2w^2w'$. Therefore (58) follows without dividing by w'. This also covers the apsides and circular-orbit limit when $j \neq 0$.

Source and reproducibility

This note reorganizes the derivations in *Space Emanation Theory - Integrated Rough Draft 6*, Sections 3.1-3.4 and 5.1-5.4, with the endpoint details from Appendix K. The source inputs, energy functional and numerical benchmark are unchanged. The intermediate algebra has been checked independently for this presentation.

Read Integrated Rough Draft 6 and its full references.